



Session 03: Density, Percent Composition & Empirical Formulas

Topic: Using numbers to identify substances and determine chemical formulas.

Time: 150 minutes

Big Fat Notebook: Chapters 3, 6

Opening Attack

Try this:

A shiny metal cube has edge length **2.00 cm** and mass **42.0 g**.

The density of aluminum is 2.70 g/cm^3 . The density of titanium is 4.51 g/cm^3 .

Is the cube aluminum or titanium? Prove it with a calculation.

A compound contains **40.0% carbon, 6.71% hydrogen, and 53.3% oxygen** by mass.

What is its simplest (empirical) formula?

Stuck? Good. By the end of this session, you'll identify unknown substances from density and derive chemical formulas from nothing but percent composition.

Worked Pattern

Example 1 — Density from Mass and Volume

Problem: A metal cube has edge 2.00 cm and mass 42.0 g. Calculate its density.

Step 1: Calculate volume.

$$\text{Volume of cube} = (\text{edge})^3 = (2.00 \text{ cm})^3 = 8.00 \text{ cm}^3$$

Step 2: Divide mass by volume.

$$\rho = \frac{\text{mass}}{\text{volume}} = \frac{42.0 \text{ g}}{8.00 \text{ cm}^3}$$

Step 3: Calculate and apply sig figs.

$$42.0 \div 8.00 = 5.25 \text{ g/cm}^3$$

Sig figs: 42.0 (3) and 8.00 (3) → 3 sig figs → 5.25 g/cm³

Answer: 5.25 g/cm³.

Compare: aluminum is 2.70, titanium is 4.51. This metal is denser than both — it's neither! (It could be germanium at 5.32 g/cm³.)

Example 2 — Identifying an Unknown by Density

Problem: An irregular solid displaces water from 25.0 mL to 28.5 mL. Its mass is 28.7 g. Is it quartz ($\rho = 2.65 \text{ g/cm}^3$) or fluorite ($\rho = 3.18 \text{ g/cm}^3$)?

Step 1: Volume by displacement.

$$V = 28.5 \text{ mL} - 25.0 \text{ mL} = 3.5 \text{ mL}$$

Remember: 1 mL = 1 cm³, so $V = 3.5 \text{ cm}^3$.

Step 2: Calculate density.

$$\rho = 28.7 \text{ g} \div 3.5 \text{ cm}^3 = 8.2 \text{ g/cm}^3$$

Step 3: Compare to known densities.

2.65 g/cm³ (quartz) — too low. 3.18 g/cm³ (fluorite) — also too low.

8.2 matches neither. (8.2 has 2 sig figs, from 3.5.)

Answer: 8.2 g/cm^3 — neither quartz nor fluorite. The calculation is right — the unknown is something else.

Example 3 — Mass Percent from a Formula

Problem: Calculate the mass percent of each element in H_2O (water).

Step 1: Find molar mass of the compound.

$$\begin{array}{rcl} & > \text{H: } 2 \times 1.008 = 2.016 \text{ g/mol} & \\ > & > \text{O: } 1 \times 16.00 = 16.00 \text{ g/mol} & > \\ & \underline{> \text{Total: } 2.016 + 16.00 = 18.016 \text{ g/mol} >} & \end{array}$$

Step 2: Mass percent of each element = (mass of element in 1 mol / molar mass) $\times$ 100%.

$$\%H = \frac{2.016}{18.016} \times 100\% = 11.19\%$$

$$\%O = \frac{16.00}{18.016} \times 100\% = 88.81\%$$

Step 3: Verify: $11.19\% + 88.81\% = 100.00\%$. ✓

Answer: $11.19\% \text{ H, } 88.81\% \text{ O by mass}$.

Example 4 — Mass Percent: Glucose

Problem: Calculate the mass percent of each element in $C_6H_{12}O_6$ (glucose).

Step 1: Molar mass.

$$\begin{array}{rcl} & > \text{C: } 6 \times 12.01 = 72.06 \text{ g/mol} & \\ > & > \text{H: } 12 \times 1.008 = 12.096 \text{ g/mol} & \\ & > \text{O: } 6 \times 16.00 = 96.00 \text{ g/mol} & > \\ & \underline{> \text{Total: } 72.06 + 12.096 + 96.00 = 180.156 \text{ g/mol} >} & \end{array}$$

Step 2: Percent of each.

$$\%C = (72.06/180.156) \times 100\% = 40.00\%$$

$$\%H = (12.096/180.156) \times 100\% = 6.714\%$$

$$\%O = (96.00/180.156) \times 100\% = 53.29\%$$

Step 3: Sum check: $40.00 + 6.714 + 53.29 = 100.00\%$. ✓

Answer: 40.00% C, 6.71% H, 53.3% O by mass (4 sig figs throughout).

Notice: This is very close to the opening problem's 40.0% C, 6.71% H, 53.3% O.
Interesting...

Example 5 — Empirical Formula from Mass Percent

Problem: A compound is 40.0% C, 6.71% H, 53.3% O by mass. Find the empirical formula.

Step 1: Assume exactly 100.0 g of the compound.

C: 40.0 g H: 6.71 g O: 53.3 g

Step 2: Convert each mass to moles by dividing by molar mass.

$$\text{mol C} = 40.0 \text{ g} \div 12.01 \text{ g/mol} = 3.3305... \text{ mol}$$

$$\text{mol H} = 6.71 \text{ g} \div 1.008 \text{ g/mol} = 6.6567... \text{ mol}$$

$$\text{mol O} = 53.3 \text{ g} \div 16.00 \text{ g/mol} = 3.33125... \text{ mol}$$

Step 3: Divide all mole values by the smallest mole value.

Smallest: ~3.33 (both C and O are close)

$$\text{C: } 3.3305/3.3305 = 1.00$$

$$\text{H: } 6.6567/3.3305 = 2.00$$

$$\text{O: } 3.3313/3.3305 = 1.00$$

Ratio: C : H : O = 1 : 2 : 1

Answer: $\boxed{CH_2O}$ (empirical formula).

This is the simplest whole-number ratio. The molecular formula could be CH_2O , $C_2H_4O_2$, $C_3H_6O_3$, etc.

Example 6 — Empirical Formula: When the Ratio Isn't Integer

Problem: A compound is 69.6% Ba, 6.09% C, 24.3% O. Find the empirical formula.

Step 1: Assume 100.0 g.

$$\text{mol Ba} = 69.6 \div 137.33 = 0.5068 \text{ mol}$$

$$\text{mol C} = 6.09 \div 12.01 = 0.5071 \text{ mol}$$

$$\text{mol O} = 24.3 \div 16.00 = 1.5188 \text{ mol}$$

Step 2: Divide by smallest (0.5068).

$$\text{Ba: } 0.5068/0.5068 = 1.00$$

$$\text{C: } 0.5071/0.5068 = 1.00$$

$$\text{O: } 1.5188/0.5068 = 3.00$$

Step 3: All are integers. Ratio = 1 : 1 : 3.

Answer: $\boxed{BaCO_3}$ (barium carbonate).

Example 7 — Empirical Formula: When You Get a .5 (Multiply by 2)

Problem: A compound is 30.4% N and 69.6% O. Find the empirical formula.

Step 1: Assume 100.0 g.

$$\text{mol N} = 30.4 \div 14.01 = 2.170 \text{ mol}$$

$$\text{mol O} = 69.6 \div 16.00 = 4.350 \text{ mol}$$

Step 2: Divide by smallest (2.170).

$$\text{N: } 2.170/2.170 = 1.00$$

$$\text{O: } 4.350/2.170 = 2.00$$

Step 3: Integers! Ratio = 1 : 2.

Answer: $\boxed{\text{NO}_2}$ (nitrogen dioxide).

Example 8 — Molecular Formula from Empirical Formula + Molar Mass

Problem: A compound has empirical formula CH_2O and molar mass 180.2 g/mol. What is the molecular formula?

Step 1: Calculate the empirical formula mass.

$$\begin{array}{rcl} & > \text{C: } 12.01 & \\ & > \text{H: } 2 \times 1.008 = 2.016 & \\ & > \text{O: } 16.00 & \\ \hline > \text{CH}_2\text{O mass: } 12.01 + 2.016 + 16.00 = 30.026 \text{ g/mol} > \end{array}$$

Step 2: Divide molecular mass by empirical mass.

$$n = 180.2 \div 30.026 = 6.00$$

Step 3: Multiply all subscripts in the empirical formula by n .

$$\text{C}_1\text{H}_2\text{O}_1 \times 6 = \text{C}_6\text{H}_{12}\text{O}_6$$

Answer: $\boxed{\text{C}_6\text{H}_{12}\text{O}_6}$ (glucose).

THE TRAP — Forgetting to Divide by the Smallest

The wrong way:

A compound is 75.0% C and 25.0% H.

Wrong work: Assume 100 g. C: $75.0/12.01 = 6.24$ mol. H: $25.0/1.008 = 24.8$ mol.

"So the formula is $C_{6.24}H_{24.8}$ " ← **WRONG!**

Why is this wrong? Chemical formulas use whole numbers. You must divide all mole values by the smallest before you can read the ratio.

The right way:

Step 1: C: $75.0 \text{ g} \rightarrow 75.0 / 12.01 = 6.245$ mol. H: $25.0 \text{ g} \rightarrow 25.0 / 1.008 = 24.80$ mol.

Step 2: Divide by smallest (6.245):

C: $6.245/6.245 = 1.00$

H: $24.80/6.245 = 3.97 \approx 4$

Formula: CH_4 (methane)

Spot the difference: Raw mole values mean nothing. The ratio AFTER dividing by the smallest gives the subscripts.

What We Just Did — The Empirical Formula Procedure

Step 1: Convert mass percent to grams by assuming exactly 100.0 g of compound.

"40.0% C" means 40.0 g of C in every 100.0 g of compound.

Step 2: Convert each mass to moles by dividing by the element's molar mass.

$\text{mol} = \text{grams} \div (\text{g/mol from the periodic table})$

Keep at least 4 decimal places — do NOT round yet.

Step 3: Divide every mole value by the smallest mole value.

You now have a ratio of small whole numbers.

If you get **.5**, multiply ALL numbers by 2.

If you get **.33** or **.67**, multiply ALL numbers by 3.

These whole numbers become the subscripts in the empirical formula.

Copy this. This 3-step procedure works for every empirical formula problem you will ever see.

Pro Tips — Calculator Shortcuts & Mental Checks

Tip 1 — The "divide by smallest" step is non-negotiable. Raw mole values are never the final subscripts. Always divide. If you skip this step, your formula will be wrong 100% of the time.

Tip 2 — Handle .5, .33, .67 with multiplication, not rounding. If after dividing you get 1 : 1.5, multiply BOTH by 2 to get 2 : 3 — not 1 : 2. If you get 1 : 1.33, multiply by 3 to get 3 : 4. Never round 1.5 to 2 or 1.33 to 1.

Tip 3 — Percent sum must equal 100% (or very close). If the given percents add to 97% or 103%, there's either an unlisted element, rounding, or experimental error. On exams, they should sum to 100.00%.

Tip 4 — Empirical $\neq$ molecular. CH_2O could be formaldehyde (CH_2O), acetic acid ($C_2H_4O_2$), lactic acid ($C_3H_6O_3$), or glucose ($C_6H_{12}O_6$). You need the molar mass to tell which one it is.

Tip 5 — Density identification trick. Calculate density $\rightarrow$ compare to list $\rightarrow$ if it doesn't match ANY listed value, don't force a match. Say "none of the above." Many students lose points by forcing a near-match (e.g., calling 5.25 "close enough" to 4.51). Density is precise — a mismatch is a mismatch.

Basic Drills (B1–B10)

Goal: Automate density and percent composition calculations.

B1. A metal block has mass 85.0 g and volume 10.0 cm³. Calculate density.

B2. A liquid has mass 34.5 g and volume 25.0 mL. Calculate density in g/mL.

B3. An object sinks in water (density 1.00 g/cm³) but floats in mercury (density 13.6 g/cm³). Its mass is 45.0 g and volume is 5.00 cm³. Calculate its density and explain why it behaves this way.

B4. Calculate the mass percent of each element in $NaCl$ (sodium chloride).

B5. Calculate the mass percent of each element in CO_2 (carbon dioxide).

B6. A compound is 52.2% C, 13.0% H, 34.8% O. Find the empirical formula.

B7. A compound is 85.6% C and 14.4% H. Find the empirical formula.

B8. A compound has empirical formula NO_2 and molar mass 92.0 g/mol. Find the molecular

formula.

B9. (Two-part) Part 1: A compound is 92.3% C and 7.7% H. Find the empirical formula. Part 2: Its molar mass is 78.1 g/mol. Find the molecular formula.

B10. (Speed round — under 2 min) A ring has mass 12.5 g and raises the water level from 20.0 mL to 21.3 mL. (a) Calculate density. (b) Is it pure silver (10.5 g/cm³) or pure gold (19.3 g/cm³)?

Advanced Drills (A1–A10)

Goal: Multi-step integration with Sessions 01–02. Exam traps.

A1. (Multi-step with Session 02) A cylinder of unknown metal has radius 1.50 cm, height 5.00 cm, and mass 250.0 g. Volume = $\pi r^2 h$. (a) Calculate volume in cm³. (b) Calculate density. (c) Use the density to identify the metal from this list: aluminum (2.70), iron (7.87), copper (8.96), lead (11.3 g/cm³).

A2. (Multi-step with Session 01) Three trials measure the mass and volume of an unknown liquid:

Trial	Mass (g)	Volume (mL)
1	14.20	18.0
2	15.05	19.0
3	13.88	17.5

Calculate density for each trial with correct sig figs. Then calculate the average density. Is this liquid water (1.00 g/mL), ethanol (0.789 g/mL), or something else?

A3. (Trap-laden) A compound is found to be 30.4% nitrogen and 69.6% oxygen. A student writes: "N: 30.4/14.01 = 2.17. O: 69.6/16.00 = 4.35. Ratio is 2.17:4.35. Multiply by 2 → 4.34:8.70. Still not integers." What did the student do wrong? Solve correctly.

A4. (Reverse engineering) A student calculates the empirical formula of a compound from: C = 40.0%, H = 6.7%, O = 53.3%. Their answer is $C_3H_6O_3$. Their partner says the empirical formula is CH_2O . Who is right? Explain the mistake.

A5. (Constructive) Write an exam problem where a compound's empirical formula and molecular formula are the SAME. (Hint: the molar mass must equal the empirical formula mass.) Solve it.

A6. (Constructive) Write a problem where the "divide by smallest" step gives a ratio ending in .33 or .67 (requiring multiplication by 3). Solve it and show why multiplication by 3 is needed.

A7. (Data table) Four unknown metals are tested:

Metal	Mass (g)	Volume (cm ³)
A	47.2	17.5
B	33.8	3.00
C	25.0	2.78
D	19.6	2.50

Known densities: magnesium (1.74), aluminum (2.70), zinc (7.13), copper (8.96), lead (11.3), gold (19.3 g/cm³). Identify each metal. Show all calculations with correct sig figs.

A8. (Timed — under 3 min) A compound contains only carbon, hydrogen, and oxygen. Combustion of 1.000 g of the compound produces 1.466 g of CO_2 and 0.600 g of H_2O . (a) Find the mass of C in the original sample (all C ends up in CO_2). (b) Find the mass of H (all H ends up in H_2O). (c) Find the mass of O by difference. (d) Determine the empirical formula.

A9. (Hardest variation) A hydrated salt has the formula $MgSO_4 \cdot xH_2O$. When 5.00 g of the hydrate is heated, 2.44 g of water is driven off. (a) Find the mass of anhydrous $MgSO_4$ remaining. (b) Convert both masses to moles. (c) Find the ratio $x = \text{mol } H_2O / \text{mol } MgSO_4$. (d) Write the full formula of the hydrate.

A10. (Exam boss) A 2.500 g sample of a compound containing only C, H, and O is burned in excess oxygen, producing 5.713 g CO_2 and 2.339 g H_2O . Separately, the molar mass is found to be 116.2 g/mol. (a) Determine the empirical formula. (b) Determine the molecular formula. (c) Calculate the mass percent of each element from the molecular formula and verify it matches the combustion data. Use correct sig figs throughout.

Now Read

You can now calculate density to identify unknown substances, derive mass percent from formulas, and determine empirical and molecular formulas from percent composition — at exam speed.

Read **Chapters 3 and 6** of *Everything You Need to Ace Chemistry in One Big Fat Notebook*.

The concepts — what density means physically, how percent composition connects to formulas, why the mole is central — will lock into place. You've solved 20 problems harder than anything the book asks.

Solutions: [solutions/03-solutions.md](https://www.khanacademy.org/a/solutions-03-solutions-md)



Key Terms — Quick Reference

Term	What it means
density	$\rho = m/V$; characteristic property used to identify substances
water displacement	$V_{\text{object}} = V_{\text{final}} - V_{\text{initial}}$; used for irregular solids
percent composition	(mass of element in 1 mol $\div$ molar mass) $\times$ 100%
empirical formula	simplest whole-number atom ratio (e.g., CH_2O , not $C_6H_{12}O_6$)
molecular formula	actual number of atoms per molecule ($n \times$ empirical formula)
$n =$ $M_{\text{molecular}}/M_{\text{empirical}}$	the multiplier to go from empirical $\rightarrow$ molecular
hydrate	ionic compound with water trapped in crystal ($MgSO_4 \cdot 7H_2O$)
anhydrous	the salt after water is driven off by heating
divide by smallest	the critical step in empirical formulas — raw mole values are NOT subscripts

Term	What it means
.5 → ×2, .33/.67 → ×3	when the ratio isn't integer, multiply ALL numbers by 2 or 3